

TIC 290129286

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

|                                 |                                 |                               |          |
|---------------------------------|---------------------------------|-------------------------------|----------|
| BLS EPHEMERIS                   |                                 | CATALOG CROSS-MATCH (novelty) |          |
| Period                          | 0.889020 d (ALIASED, see below) | VSX                           | no match |
| -> corrected period: 1.778041 d |                                 | ASAS-SN                       | no match |
| Epoch (t0)                      | 1325.7791 BTJD                  | Gaia eclipsing-binary table   | no match |
| Duration                        | 0.0840 d                        | Gaia variable classification  | no match |
| Depth                           | 15.564 % (155,641 ppm)          | TESS EB catalog (Prsa+2022)   | no match |
| SNR                             | 256.7                           | ExoFOP TOI list               | no match |
| INDEPENDENT CHECKS (f3i gates)  |                                 | Verdict: NOVEL                |          |
| Eclipses observed               | 26 (gate >= 3)                  |                               |          |
| Robust depth ratio              | 0.86 (gate >= 0.50)             |                               |          |
| Robust depth sigma              | 54.9 (gate >= 7.0)              |                               |          |
| Odd/even z-score                | 12.38 (ALIAS FLAGGED)           |                               |          |
| Secondary eclipse               | not detected                    |                               |          |

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

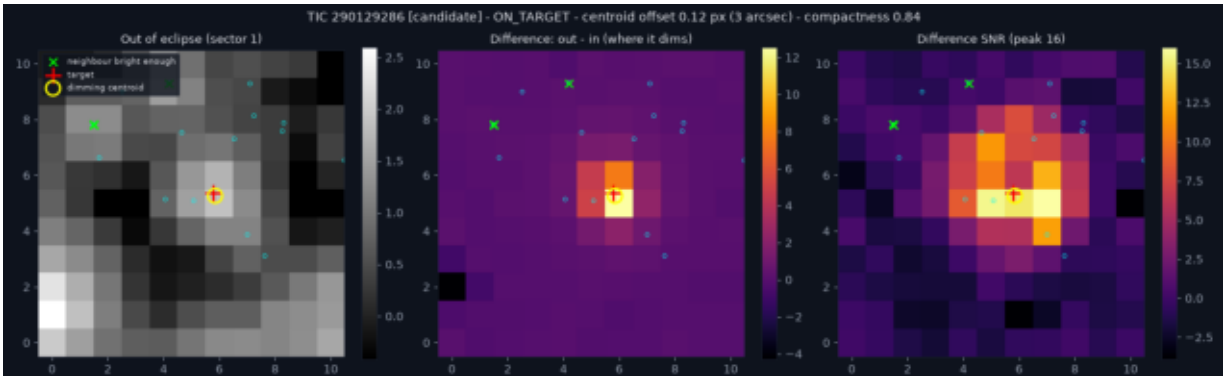
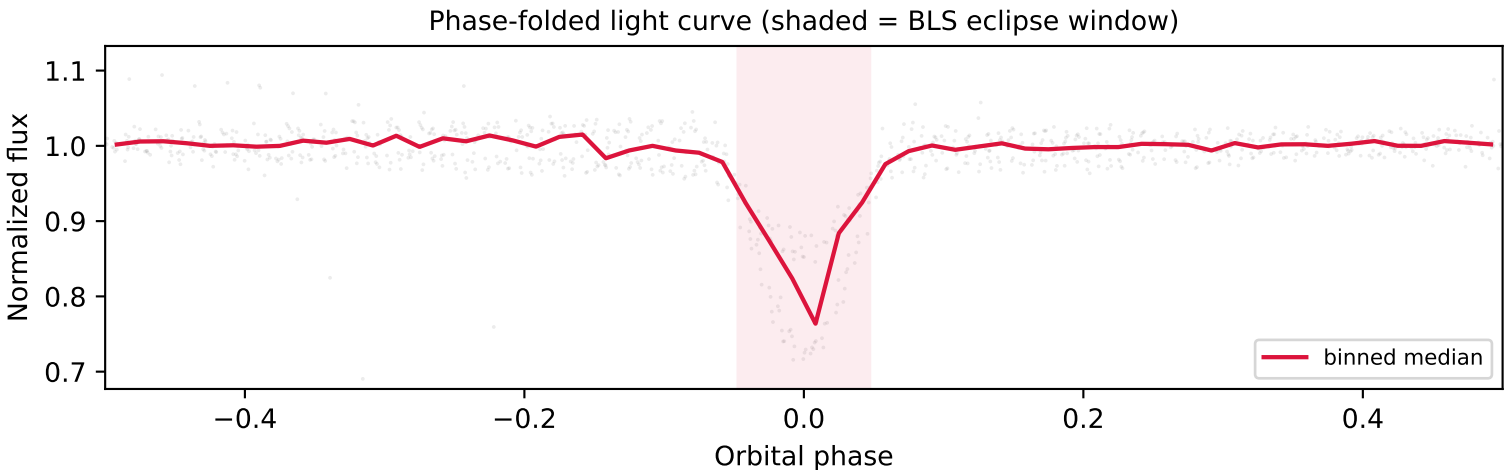
Sector 1 light-curve dip in this sector: 47.0 sigma t0 refined by 0.0025 phase

Difference-image peak SNR 16.05 (gate >= 5.0) Compactness 0.84 (gate >= 0.50)

Dimming centroid offset 0.12 px (3 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 290129279

4.38 px away, dTmag +1.09 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 12.38 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 1.778041 d, in any write-up.